# THE FUNDAMENTAL TUG-OF-WAR IN PHYSICAL AI

Reconciling High-Capacity Stochastic Deliberation with Deterministic Real-Time Physics

## AI DELIBERATION REALM

### Cognitive Capacity

Demands Compute, Tokens & Time

**Computational Nature:**

- High-capacity VLMs & Diffusion policies
- Billions of parameters & KV cache matrices
- Non-deterministic stochastic outputs

**Temporal Profile:**

**`Slow Cadence: 0.5–50 Hz`**

**`P99 Latency Tails ≈ 500–1500 ms`**

Operating substrate: Linux MPU / Edge NPU.

## STRUCTURAL TENSION

Latent Time vs. Physical Reality

## PHYSICAL DYNAMICS REALM

### Kinetic Determinism

Demands Microseconds & Hard Bounds

**Physical Reality:**

- Kinetic momentum $p = mv$ & inertia
- Dynamic stopping distance $d_{stop}(v)$
- Actuator Joule heating & gear shear

**Temporal Constraint:**

**`Fast Reflex: 1000 Hz (1.0 ms)`**

**`Hard Deadline: τ_world ≤ 1–10 ms`**

Operating substrate: Real-Time Bare-Metal MCU.

## HOW PHYSICAL AI RECONCILES THE TUG-OF-WAR: THREE ARCHITECTURAL CONTRACTS

### 1. Proposal–Permission Privilege Split

- Untrusted MPU proposals never touch motors.
- Dedicated 1 kHz MCU filters commands via Control Barrier Functions: $h(x) \geq 0$.

### 2. Action Chunking (System 1.5)

- Amortizes neural latency over future horizon H.
- Synthesizes continuous $C^2$ jerk-bounded trajectories executed locally at 1 kHz.

### 3. Expiring Intent Leases ($L_{intent}$)

- Binds proposals to time-to-live validity (TTL).
- Continuous state tracker revokes lease on drift, commanding deterministic safe position hold.